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Experimental Research: Cause and Effect Relationships

Experiment

Direct manipulation of a variable to see if corresponding changes in behaviour result, allowing the determination of cause and effect relationships

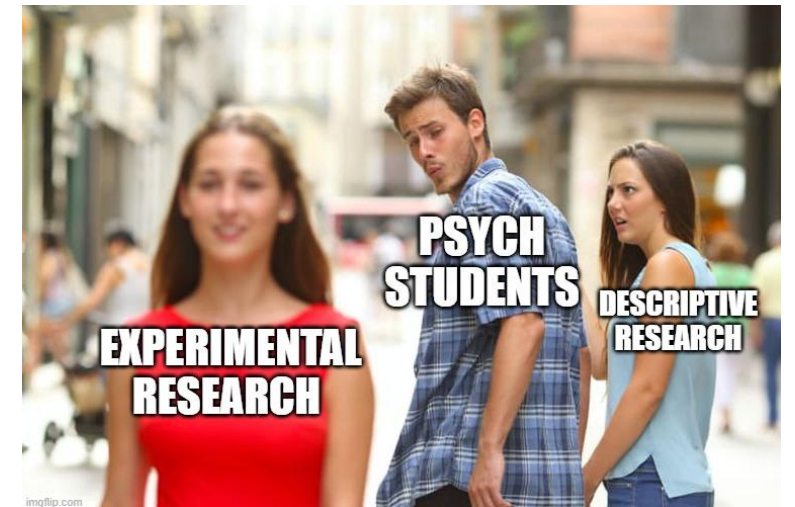


Operational definition

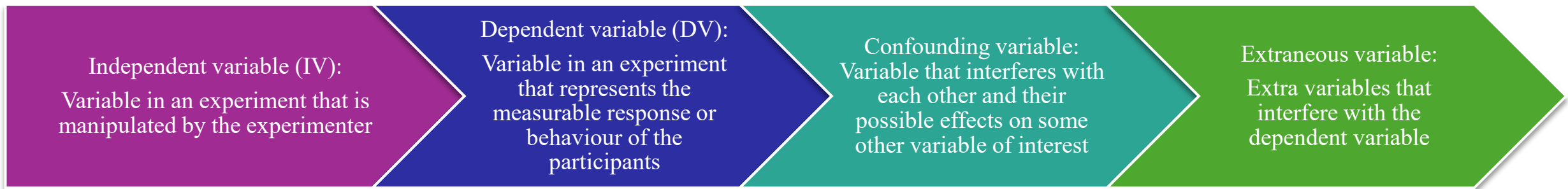
Definition of a variable of interest that allows it to be directly measured; specific

Variable

Anything that can be measured, manipulated, or controlled

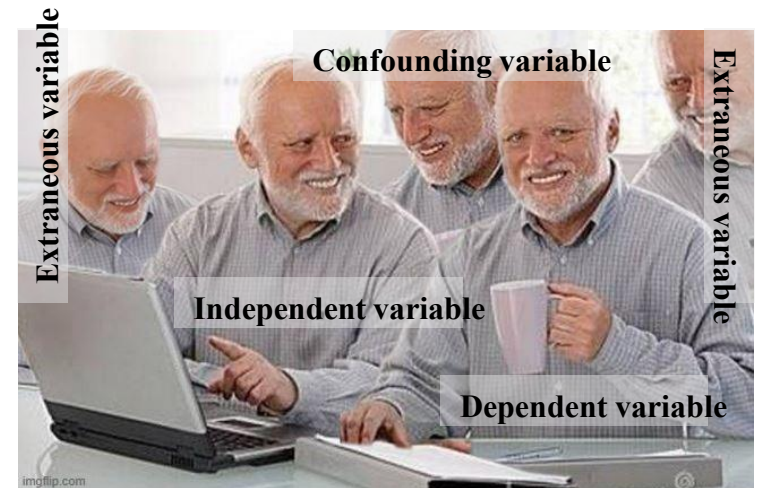


Experimental Research: Cause and Effect Relationships (cont'd...)



Experimental group: independent variable (receives the experimental manipulation)

Control group: participants in an experiment who are not subjected to the independent variable and who may receive a placebo treatment



Cause and Effect Relationships: Experimental Research (cont'd...)

Section	Details
 Title	The Impact of Social Media Use on Self-Esteem in Gen Z College Students
 Research Question	Does limiting social media usage improve self-esteem in Gen Z college students?
 Hypothesis	Limiting social media to 30 min/day for 2 weeks will increase self-esteem compared to usual usage
 Independent Variable	- Daily social media use: - Experimental: ≤ 30 min/day - Control: usual use
 Dependent Variable	Self-esteem measured by Rosenberg Self-Esteem Scale before and after two weeks
 Groups	- Experimental: limit use (monitored) - Control: no change
 Confounding Variables	- Mental health, life events, personality, other digital activities - Controlled via screening and design
 Extraneous Variables	- Testing environment, time of day, mood, platform type - Minimised by standardising conditions
 Design Summary	Random assignment, pre/post-tests, compliance monitoring, 2-week duration



Independent Variable (IV) :

Two potion ingredients: 1. Jobberknoll feather, 2. Bicorn horn

Dependent Variable (DV):

Transparent potion

Control Variable (CV):

Other ingredients

Hermione is trying to prepare a veritas serum. However, she is unable to recollect all the ingredients. She is mixed between Jobberknoll feather and Bicorn horn.

One way to figure out this is by trying to add each of the ingredients and check which one works.

Adding the correct ingredient will make the potion make it transparent.

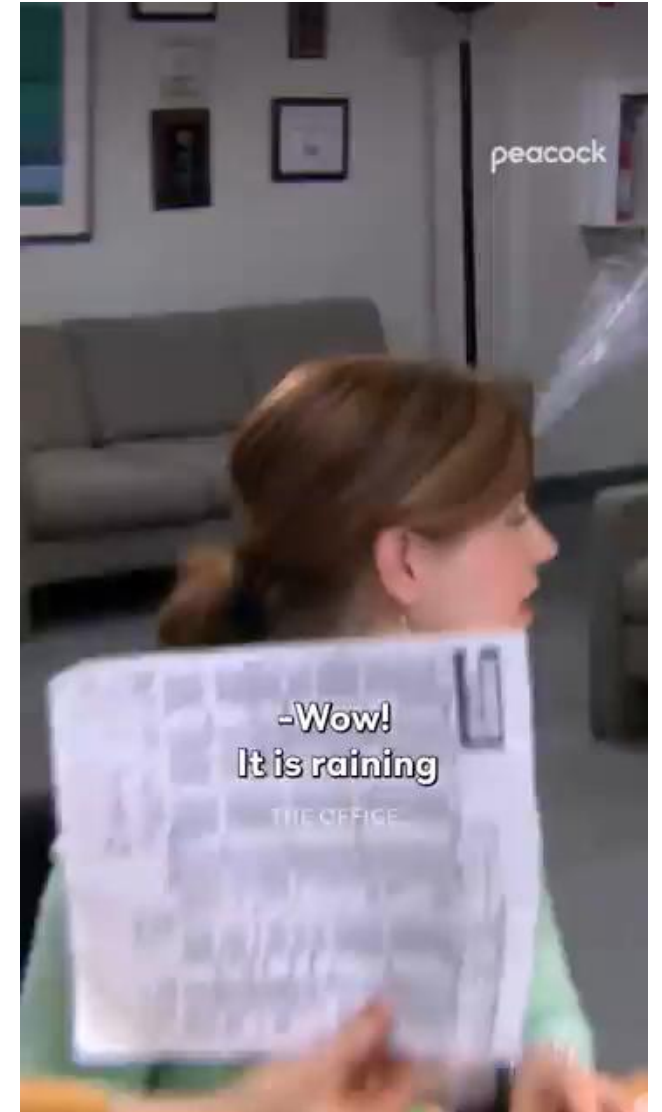


1. Go to [wooclap.com](https://www.wooclap.com)
2. Enter the event code in the top banner
Event code
[FYPMEX](#)

Experimental Research: Drawbacks (cont'd...)

Experimenter effect:

tendency of the experimenter's expectations for a study to unintentionally influence the results of the study



The Office

Cause and Effect Relationships: Experimental Research (cont'd...)

Single-anonymous study:

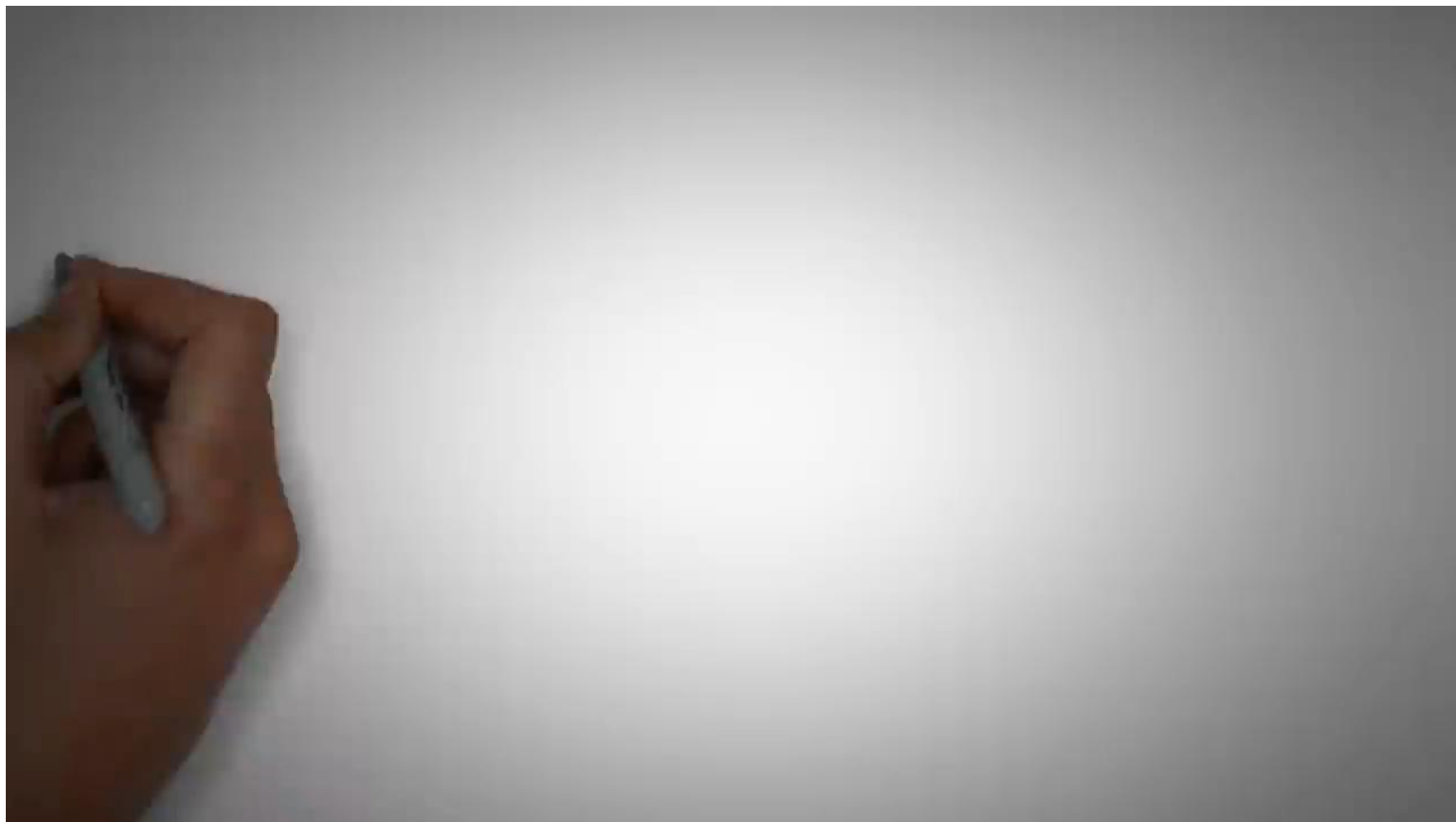
a study in which the participants do not know if they are in the experimental or the control group

Double-anonymous study:

a study in which neither the experimenter nor the participants knows if the participants are in the experimental or control group



Ethical Issues in Psychological Research



Ethical Issues in Psychological Research

- **Rights and well-being:** weighed against the study's value to science
- **Informed consent** for participation
- **Debriefing:** Deception must be justified; reveal true nature of the study and expectations of results
- **Right to withdraw** from the study at any time
- Safeguard from **potential risks/emotional distress**
- **Confidentiality**



Key Stuff to Lock In



- Hypothesis
- Clinical case studies
- Naturalistic observation
- Observer effect
- Participant observation
- Observer bias
- Surveys
- Representative sample
- Courtesy bias
- Wording Bias

- Question-order bias
- Cross-sectional research
- Longitudinal research
- Correlational research
- Experimental research
- Operational definition
- Variables
- Independent variables
- Dependent variables
- Confounding bias

- Confounding variables
- Extraneous variables
- Experimental group
- Control group
- Experimenter effect
- Single-anonymous study
- Double-anonymous study
- Informed consent
- Debriefing
- Confidentiality

References

- Ciccarelli, S. K., & Meyer, G. E. (2006). *Psychology*. Pearson Education.
- Coon, D., & Mitterer, J. O. (2010). *Introduction to psychology* (12th ed.). Wadsworth Cengage Learning.
- Feldman, R. S. (2009). *Essentials of understanding psychology* (8th ed.). McGraw-Hill.

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